

BANKING FRAUD ANALYSIS REPORT

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Executive Summary:

This report evaluates the analysis of **1,000** banking transactions to uncover fraud patterns, device vulnerabilities and operational risks. The analysis reveals a critical overall fraud rate of 48.1 %, indicating an urgent need for security intervention.

Three primary risk areas were identified:

1. Mobile Device Vulnerability:

Mobile transactions show a significantly higher fraud rate (**51.25%**) compared to Desktop (**44.6%**), making mobile platforms the primary vector for fraudulent activity.

2. High-Risk Geolocations:

Five specific geolocation coordinates were found to have a **100% fraud rate**. These locations pose an immediate threat and require instant countermeasures such as IP blacklisting

3. Anomalous Temporal Pattern:

A significant surge in transaction volume is detected at **11:00 AM**, exhibiting high-risk characteristics consistent with automated bot-attack patterns.

Project Introduction:

1. Project Title:

Banking Fraud Analysis Analysis

2. Objective:

The objective of this project was to conduct a comprehensive, end-to-end analysis of 1,000 banking transactions to diagnose fraud architectures and quantify operational risks across diverse device ecosystems and geographies. This culminated in the development of a dynamic dashboard designed to facilitate data-driven, real-time security interventions.

3. Dataset Overview:

- Source: Kaggle
- Scope: 1,000 unique banking transactions.
- Key Attributes: Transaction ID, Sender Account ID, Receiver Account ID, Transaction Amount, Transaction Type, Timestamp, Transaction Status, Fraud Flag, Geolocation (Latitude/Longitude), Device Used, Network Slice ID, Latency (ms), Slice Bandwidth (Mbps), PIN Code.

Tools Used:

- MySQL Workbench: ETL, KPI aggregation, and the creation of optimized summary tables for risk and performance metrics.
- Google Sheets: Development of an interactive dashboard featuring automated watchlists and real- time KPI tracking.

Methodology:

1. Data Inspection

- a. Total transactions: 1,000
- b. Transaction Types: Deposit, Withdrawal, Transfer
- c. Device Type: Desktop, Mobile
- d. Geolocations: Multiple coordinates (Latitude/Longitude)
- e. Transaction Status: Failed, Success
- f. Fraud Flag: True, False

2. Data Validation

- Checked for consistency in data types across all columns (e.g., Transaction Amount as numeric, Timestamp as datetime).
- Validated referential integrity between transaction table.

3. Data Cleaning Strategy

- a. No duplicates found
- b. No missing values
- c. Standardized date format

4. Data Aggregation (SQL)

- **Segmented Fraud Risk:** Performed complex grouping by device type to calculate specific fraud percentages, identifying a 51.25% risk on mobile platforms.
- **Geospatial Clustering:** Aggregated transaction counts and fraud flags by coordinate to isolate the top 5 high-threat geolocations.
- **Temporal Analysis:** Extracted hourly patterns from timestamps to identify the 11:00 AM surge and quantify bot-attack characteristics.
- **Performance Metrics:** Calculated transaction success and failure rates by category (Deposit, Withdrawal, Transfer) using aggregate functions.

5. KPI Calculations

- a. KPI 1: Total Transactions = 1,000
- b. KPI 2: Total Volume Processed = \$ 771,165.29
- c. KPI 3: Average Transaction Value = \$ 771.17
- d. KPI 4: Total Fraud Cases = \$ 481
- e. KPI 5: Overall Fraud Rate = 48.1%

6. Exploratory Analysis

a. Fraud Risk by Device Type

Device_Used	Total_Transactions	Fraud_Cases	Fraud_Rate_Percentage
Desktop	479	214	44.68%
Mobile	521	267	51.25%

b. High-Risk Geolocation Watchlist

Geolocation	Total_Transactions	Fraud_Count
51.5074 N, -118.2437 W	21	21
34.0522 N, 0.1278 W	20	20
48.8566 N, 139.6917 W	20	20
51.5074 N, 0.1278 W	20	20
55.7558 N, 2.3522 W	19	19

c. Temporal Fraud Pattern

Hour_of_Day	Total_Transaction	Fraud_Cases	Fraud_Rate
10:00 AM	983	472	48.02
11:00 AM	17	9	52.94

d. Transaction Type and Status Performance

Transactions_Type	Transaction_Status	Transaction_Count
Deposit	Failed	166
Withdrawal	Success	145
Withdrawal	Failed	165
Transfer	Success	192
Deposit	Success	150
Transfer	Failed	182

7. Data Migration

- **Export Process:** Cleaned and aggregated datasets were exported from MySQL Workbench as standardized CSV files.
- **Data Integrity Check:** Verified that row counts and calculated values matched perfectly between the SQL output and the imported Sheets data.
- **Frontend Integration:** Imported the processed CSV files into Google Sheets to serve as a structured backend for the dashboard interface.

8. Dashboard Development

Built an -

- KPI Summary Cards
- Transaction Distribution by Status Chart
- Fraud Risk by Device Type Chart
- Hourly Fraud Pattern Analysis Chart
- High - Risk Geolocation Watchlist
- Pro - Active Risk Summary



Key Insights:

1. Core Finding: Systemic Risk Exposure

Analysis reveals a critical overall fraud rate of **48.1%**, signaling widespread systemic gaps across the banking ecosystem. The scale of these findings suggests that existing controls are insufficient to mitigate current threat actors, making the deployment of enhanced security measures a high-priority operational requirement.

2. Primary Vulnerability Vector: Mobile Ecosystems

Data Analysis identifies mobile platforms as the primary vulnerability vector, exhibiting a disproportionate fraud of **51.25%** compared to desktop environments. This trend indicates targeted exploitation of mobile-specific security frameworks, potentially involving SMS-based credential interception or unauthorized application-layer access.

3. Geospatial Integrity: High-Threat Clusters

Spatial analysis detected five distinct geolocation coordinates with a 100% fraud saturation rate. The absolute correlation between these locations and fraudulent activity indicates the presence of compromised network nodes or organized fraud syndicates. Consequently, these high-threat zones have been flagged for immediate systemic blacklisting to preserve the integrity of the broader transaction ecosystem.

4. Short-Burst Automated Attacks

The report identifies a critical temporal vulnerability at 11:00 AM, characterized by a **52.94%** fraud rate despite low overall traffic. This pattern is consistent with coordinated bot activity, where attackers utilize short, high-intensity bursts to validate compromised credentials without triggering volume-based security thresholds.

5. Uniform Risk Distribution Across Transaction Modalities

Analysis reveals that fraudulent activity is distributed almost evenly across all primary transaction types: Transfers (182), Deposits (166), and Withdrawals (165). This balanced risk profile vulnerability. Consequently, the findings necessitate a holistic security architecture that provides end-to-end protection across the entire transaction lifecycle.

Growth Opportunities & Recommendations:

Strategic Recommendations for Enhanced Operational Security:

1. Enforce mobile-specific multi-factor authentication (MFA) to address the 51.25% fraud rate on mobile devices.
2. Immediately blacklist the five identified geolocations exhibiting a 100% fraud rate to block compromised nodes.
3. Conduct a forensic investigation of the 11:00 AM transaction surge to determine if automated bot-attack patterns are present.

4. Deploy real-time alerting for failed transaction clusters (Deposits, Withdrawals, Transfers) to enable proactive fraud prevention.
5. Implement dynamic risk scoring based on device type, geolocation, and time of day for automated transaction blocking or step-up authentication.

Appendix: Technical Implementation (SQL Scripts)

1. Database Schema & Data Initialization

```
CREATE DATABASE BankingAnalysis;
USE BankingAnalysis;

CREATE TABLE Transactions (
    Transaction_ID VARCHAR(255),
    Sender_Account_ID VARCHAR(255),
    Receiver_Account_ID VARCHAR(255),
    Transaction_Amount VARCHAR(255),
    Transaction_Type VARCHAR(255),
    Timestamp VARCHAR(255),
    Transaction_Status VARCHAR(255),
    Fraud_Flag VARCHAR(255),
    Geolocation VARCHAR(255),
    Device_Used VARCHAR(255),
    Network_Slice_ID VARCHAR(255),
    Latency_ms VARCHAR(255),
    Slice_Bandwidth_Mbps VARCHAR(255),
    PIN_Code VARCHAR(255)
);
```

2. Data Cleaning & Type Optimization

```
SET SQL_SAFE_UPDATES = 0;
UPDATE Transactions
SET Fraud_Flag = CASE
    WHEN Fraud_Flag = 'TRUE' THEN '1'
    WHEN Fraud_Flag = 'FALSE' THEN '0'
    ELSE Fraud_Flag
END;
SET SQL_SAFE_UPDATES = 1;
```

```

ALTER TABLE Transactions MODIFY COLUMN Transaction_Amount
DECIMAL(15,2);
ALTER TABLE Transactions MODIFY COLUMN Timestamp DATETIME;
ALTER TABLE Transactions MODIFY COLUMN Latency_ms INT;
ALTER TABLE Transactions MODIFY COLUMN Slice_Bandwidth_Mbps INT;
ALTER TABLE Transactions MODIFY COLUMN PIN_Code INT;
ALTER TABLE Transactions MODIFY COLUMN Fraud_Flag TINYINT(1);

```

3. Analytical Queries for Dashboard Aggregation

I. Device-Based Risk

```

SELECT
    Device_Used,
    COUNT(*) AS Total_Transactions,
    SUM(Fraud_Flag) AS Total_Fraud_Cases,
    ROUND((SUM(Fraud_Flag) / COUNT(*)) * 100, 2) AS
    Fraud_Rate_Percentage
FROM Transactions
GROUP BY Device_Used
ORDER BY Fraud_Rate_Percentage DESC;

```

II. High-Risk Geolocation

```

SELECT
    Geolocation,
    COUNT(*) AS Total_Transactions,
    SUM(Fraud_Flag) AS Fraud_Count
FROM Transactions
WHERE Fraud_Flag = 1
GROUP BY Geolocation
ORDER BY Fraud_Count DESC
LIMIT 5;

```

III. Hourly Transaction Trends

```

SELECT
    HOUR(Timestamp) AS Hour_of_Day,
    COUNT(*) AS Total_Transactions,

```



```

SUM(Fraud_Flag) AS Fraud_Cases,
ROUND((SUM(Fraud_Flag) / COUNT(*)) * 100, 2) AS Fraud_Rate
FROM Transactions
GROUP BY Hour_of_Day
ORDER BY Hour_of_Day;

```

IV. Transaction Type & Status Performance

```

SELECT
    Transaction_Type,
    COUNT(*) AS Total_Count,
    SUM(CASE WHEN Transaction_Status = 'Success' THEN 1 ELSE 0 END) AS
Success_Count,
    SUM(CASE WHEN Transaction_Status = 'Failed' THEN 1 ELSE 0 END) AS
Failure_Count,
    ROUND((SUM(CASE WHEN Transaction_Status = 'Success' THEN 1 ELSE 0
END) / COUNT(*)) * 100, 2) AS Success_Rate_Percentage
FROM Transactions
GROUP BY Transaction_Type;

```

4. Operational Risk & Security Auditing

```

SELECT
    Sender_Account_ID,
    COUNT(*) AS Failed_Attempts
FROM Transactions
WHERE Transaction_Status = 'Failed'
GROUP BY Sender_Account_ID
HAVING Failed_Attempts > 3
ORDER BY Failed_Attempts DESC;

```